RBE/CS 549: Computer Vision Project 2 – Buildings Built in Minutes: SfM and NeRF – Phase 2 Report

Filippo Marcantoni
Robotics Engineering
Worcester Polytechnic Institute
fmarcantoni@wpi.edu

Prahladh Raja
Robotics Engineering
Worcester Polytechnic Institute
pnamboorkrishnam@wpi.edu

***Abstract*—This report details the implementation and evaluation of *NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis* as proposed by Mildenhall et al. (2020). We analyze the transition from explicit geometric representations to implicit neural scene representations. The report covers the mathematical foundations of continuous volumetric scene functions, differentiable volume rendering, and high-frequency feature recovery via positional encoding. Furthermore, we detail our PyTorch implementation, featuring a hierarchical sampling strategy utilizing coarse and fine Multi-Layer Perceptrons (MLPs). Our results demonstrate state-of-the-art view synthesis performance, evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) metrics.**

I. Introduction and Motivation

Novel view synthesis is the task of generating photorealistic images from viewpoints not present in a given set of input photographs, and has been a long-standing challenge in computer vision and computer graphics. Classical approaches typically rely on explicit 3D geometric representations, such as triangle meshes, point clouds, or discretized voxel grids. While effective in controlled settings, these representations suffer from fundamental limitations: voxel grids scale cubically in memory with resolution ($O(n^3)$), meshes require manual construction or template initialization, and none of them naturally model complex, non-Lambertian view-dependent light effects such as specular reflections or semi-transparent materials.

Neural Radiance Fields (NeRF) [1] constitute a paradigm shift in scene representation. Rather than storing geometry and appearance in an explicit data structure, NeRF encodes the entire scene implicitly in the weights of a deep Multi-Layer Perceptron (MLP). The scene is defined as a continuous 5D function:

$$F_\Theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma), \tag{1}$$

where $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$ denotes a spatial coordinate in the scene, $\mathbf{d} = (\theta, \phi) \in \mathbb{S}^2$ is the unit viewing direction, $\mathbf{c} = (r, g, b) \in [0, 1]^3$ is the view-dependent emitted radiance (color), and $\sigma \geq 0$ is the volume density at that point. The volume density $\sigma(\mathbf{x})$ acts as a differential opacity: it describes the probability that an infinitesimal ray segment passing through $\mathbf{x}$ is absorbed. Since the color $\mathbf{c}(\mathbf{x}, \mathbf{d})$ depends on both position and viewing direction, it enables the representation of view-dependent effects such as specularities and iridescence.

The key insight enabling NeRF to be trained purely from 2D images is the use of *differentiable volume rendering*. By numerically integrating the radiance field along camera rays and comparing the result against observed pixel colors, the system can be optimized end-to-end with gradient descent using only posed RGB images as supervision, with no 3D ground-truth required. Two additional innovations are critical to achieving high-quality results: (1) a Fourier *positional encoding* that lifts the low-dimensional input coordinates into a high-dimensional space, overcoming the spectral bias of MLPs toward smooth, low-frequency functions; and (2) a *hierarchical sampling* strategy that uses a coarse network to guide the allocation of more samples to high-density regions of the scene, dramatically improving rendering efficiency.

This report is organized as follows. Section II reviews related work. Section III provides a rigorous treatment of the theoretical components. Section IV describes our PyTorch implementation in detail. Section V presents our quantitative and qualitative evaluation. Section VII concludes.

II. Related Work

A. Neural 3D Scene Representations

Prior work on representing 3D scenes with neural networks falls into two broad categories. Occupancy Networks [4] and DeepSDF [5] map 3D coordinates to occupancy probabilities or signed distance values, learning compact shape representations. However, these methods require access to ground-truth 3D geometry for supervision and cannot natively model complex appearance. Scene Representation Networks (SRN) [2] take a step toward view synthesis by learning a feature-valued MLP and using a differentiable recurrent march to locate surfaces, but their single-surface model is fundamentally unable to represent volumetric phenomena such as smoke, hair, or translucency. NeRF avoids this restriction by modeling the full volume density field rather than only the scene's surface.

B. Volumetric and Image-Based Rendering

Neural Volumes [6] represent dynamic objects as voxel grids, achieving good quality but with $O(n^3)$ memory scaling at high resolutions. Local Light Field Fusion (LLFF) [7] synthesizes novel views for well-sampled forward-facing scenes at high quality but stores a separate multi-plane image (MPI) for each input view, resulting in datasets exceeding 15 GB for a single scene. NeRF requires only ≈ 5 MB for the network weights, which represents a compression factor of $3000\times$ relative to LLFF.

C. Frequency Bias and Positional Encoding

Deep MLPs demonstrated a well-documented *spectral bias* towards learning low-frequency functions and converge slowly on high-frequency ones [3]. This makes direct regression from raw Cartesian coordinates unsuitable for representing fine geometric and texture detail. The solution, borrowed from the Transformer architecture [10], is to project the inputs into a higher-dimensional space via periodic functions before passing them to the MLP. NeRF applies this idea to continuous 5D coordinates to learn high-frequency detailed textures.

III. Theoretical Background

A. Ray Generation and Camera Model

For every pixel (u, v) in a target image, we cast a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, where $t \in [t_n, t_f]$ is the depth parameter, t_n and t_f are the near and far bounds of the scene, $\mathbf{o}$ is the camera's optical center in world coordinates, and $\mathbf{d}$ is a unit direction vector.

Given the camera intrinsic matrix $\mathbf{K}$ with focal lengths f_x, f_y and principal point (c_x, c_y), the direction of the ray through pixel (u, v) is first computed in the camera frame:

$$\mathbf{d}_{\text{cam}} = \left(\frac{u - c_x}{f_x},\ \frac{v - c_y}{f_y},\ 1.0 \right), \tag{2}$$

where the z-component convention assumes the camera looks along $-z$ in camera space, meanwhile the y-component is flipped because Y increases upward in the camera space while v increases downward in the image space. The unit world-space direction is then:

$$\mathbf{d} = \frac{\mathbf{R}_{c2w}\, \hat{\mathbf{d}}_{\text{cam}}}{\|\mathbf{R}_{c2w}\, \hat{\mathbf{d}}_{\text{cam}}\|}, \tag{3}$$

where $\hat{\mathbf{d}}_{\text{cam}} = \mathbf{d}_{\text{cam}}/\|\mathbf{d}_{\text{cam}}\|$ and $\mathbf{R}_{c2w} \in SO(3)$ is the rotation sub-block of the 4×4 camera-to-world transformation matrix. The ray origin is simply the camera translation: $\mathbf{o} = \mathbf{t}_{c2w}$.

In our implementation (`PixelToRay` in `Wrapper.py`), the intrinsic matrix $\mathbf{K}$ is derived from the horizontal field-of-view angle stored in the Blender `transforms_*.json` files:

$$f = \frac{W/2}{\tan(\theta_{fov}/2)}, \tag{4}$$

where W is the image width in pixels and θ_{fov} is the camera's horizontal field of view. The principal point is assumed to be centered: $c_x = W/2$, $c_y = H/2$.

B. Differentiable Volume Rendering

The physical basis for NeRF's rendering equation is classical volume rendering [8]. Consider a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$. At any point along the ray, the radiance field provides a volume density $\sigma(\mathbf{r}(t))$ and an emitted color $\mathbf{c}(\mathbf{r}(t), \mathbf{d})$. The expected color of the ray is:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\, \sigma(\mathbf{r}(t))\, \mathbf{c}(\mathbf{r}(t), \mathbf{d})\, dt, \tag{5}$$

where the accumulated transmittance $T(t)$ represents the probability that the ray travels from t_n to t without being absorbed by any intervening particle:

$$T(t) = \exp\left(- \int_{t_n}^{t} \sigma(\mathbf{r}(s))\, ds \right). \tag{6}$$

Intuitively, $T(t)$ is a *visibility* term: if the density is high at points between t_n and t, the transmittance is low and the contribution of the color at t to the final pixel is heavily attenuated.

Because the continuous integral in Eq. (5) cannot be evaluated analytically for an arbitrary MLP, it must be approximated. The paper proposes a discrete quadrature approximation that partitions $[t_n, t_f]$ into N samples $\{t_i\}_{i=1}^{N}$ with inter-sample distances $\delta_i = t_{i+1} - t_i$:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^{N} T_i \left(1 - \exp(-\sigma_i \delta_i)\right) \mathbf{c}_i, \tag{7}$$

$$T_i = \exp\left(- \sum_{j=1}^{i-1} \sigma_j \delta_j \right). \tag{8}$$

The term $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$ is the discrete opacity for sample i, which is the probability that the ray is absorbed within the i-th segment. The product $w_i = T_i \alpha_i$ is the *rendering weight*, representing how much sample i contributes to the final composite color. Note that $\sum_i w_i \leq 1$, with equality when the ray is fully absorbed before reaching t_f.

This formulation is critical for two reasons. First, it reduces to standard alpha compositing [9], and second, every operation is differentiable with respect to σ_i and $\mathbf{c}_i$. Therefore, gradients of the pixel-level mean squared error loss flow directly back to the MLP parameters, enabling end-to-end training.

C. Positional Encoding

Although MLPs are universal function approximators in theory, their gradient-based optimization in practice is strongly biased toward learning low-frequency, smooth solutions, which is a phenomenon known as the *spectral bias* [3]. When NeRF is trained by regressing directly from 5D input coordinates (x, y, z, θ, ϕ), the resulting renderings are blurry and lack sharp edges and fine textures, as the network converges to an over-smoothed scene representation.

NeRF addresses this by passing each input coordinate through a fixed, non-learned Fourier feature mapping $\gamma(\cdot)$ before the MLP. This mapping embeds the scalar input p into a

$2L$-dimensional vector of sinusoidal features at exponentially increasing frequencies:

$$\gamma(p) = \big(\sin(2^0\pi p),\ \cos(2^0\pi p),\ \ldots,\ \sin(2^{L-1}\pi p),\ \cos(2^{L-1}\pi p)\big). \tag{9}$$

The function γ is applied independently to each component of the input. For a 3D position $\mathbf{x} = (x, y, z)$, the encoding $\gamma(\mathbf{x})$ concatenates $\gamma(x)$, $\gamma(y)$, and $\gamma(z)$, yielding a vector of dimension $3+3\times2\times L$. With $L = 10$ for positions, this gives $3 \times 2 \times 10 = 63$ dimensions. Similarly, the 3D unit direction vector $\mathbf{d}$ is encoded with $L = 4$, producing $3+3\times2\times4 = 27$ dimensions.

Geometrically, this mapping projects the inputs onto a high-dimensional manifold in the encoding space. The inner product between two encoded points captures high-frequency correlations that are invisible to the raw coordinates, allowing the MLP to represent fine-grained variation. As demonstrated in the ablation study of [1], removing positional encoding is the single largest performance degradation, confirming its central importance.

D. MLP Architecture

The full NeRF function F_Θ is decomposed into two sub-functions to enforce multiview consistency:

$$\sigma = f_\sigma(\gamma(\mathbf{x})), \qquad \mathbf{c} = f_c(\gamma(\mathbf{x}), \gamma(\mathbf{d})). \tag{10}$$

The density σ is predicted from position alone, ensuring that the geometry of the scene is view-independent. The color $\mathbf{c}$ depends on both position and direction, modeling non-Lambertian appearance.

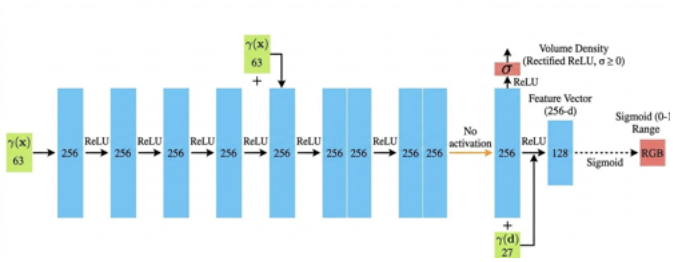

Fig. 1. MLP Architecture

Concretely, the MLP processes the encoded position $\gamma(\mathbf{x}) \in \mathbb{R}^{63}$ through 8 fully-connected ReLU layers, each with 256 hidden units. Following the DeepSDF architecture [5], a *skip connection* re-injects the positional encoding at layer 5 by concatenating it with the hidden representation, producing a $(256 + 63)$-dimensional input for that layer. This prevents the network from forgetting the original coordinates during the deep forward pass. After layer 8, two outputs are branched off:

- A scalar density σ, predicted by a single linear layer followed by a ReLU activation to enforce non-negativity.
- A 256-dimensional feature vector $\mathbf{h}$ (no activation), which is then concatenated with the encoded viewing direction $\gamma(\mathbf{d}) \in \mathbb{R}^{27}$ and processed by one additional 128-unit ReLU layer to produce the 3-channel RGB color output via a sigmoid activation.

Two separate instances of this MLP, referred to as the *coarse* and *fine* networks, are trained jointly, sharing no weights. Their architectural hyperparameters are identical; they differ only in the sampling strategy used to select their inputs.

E. Hierarchical Volume Sampling

Uniform sampling along a ray is wasteful: the majority of the ray typically passes through empty space or occluded regions that contribute negligibly to the final pixel color. The hierarchical strategy introduced in [1] allocates samples more efficiently by concentrating them near actual scene surfaces.

The procedure operates in two passes per ray:

1. Coarse Stratified Sampling: The interval $[t_n, t_f]$ is divided into $N_c = 64$ equally-spaced bins. One sample t_i is drawn uniformly at random from within each bin (*stratified sampling*):

$$t_i \sim \mathcal{U}\left[t_n + \frac{i-1}{N_c}(t_f - t_n),\ t_n + \frac{i}{N_c}(t_f - t_n)\right]. \tag{11}$$

Stratified sampling guarantees that samples are spread across the entire depth range, avoiding the clustering that would result from purely random sampling, while still introducing stochasticity that prevents the MLP from overfitting to a fixed set of query positions.

The coarse MLP is queried at these N_c points. The resulting rendering weights $w_i = T_i(1 - \exp(-\sigma_i\delta_i))$ are computed using Eq. (7).

2. Fine Importance Sampling: The coarse weights $\{w_i\}$ are normalized into a piecewise-constant probability density function (PDF) along the ray:

$$\hat{w}_i = \frac{w_i}{\sum_{j=1}^{N_c} w_j}. \tag{12}$$

A set of $N_f = 128$ additional depth samples is drawn from this PDF using *inverse transform sampling*, also known as the inverse CDF method. This technique samples proportionally to density: regions where $\hat{w}_i$ is large (i.e., where scene content is likely to exist) receive more fine samples, while empty regions receive fewer.

The $N_c + N_f = 192$ combined (and sorted) depth samples are then passed to the fine MLP, which predicts the final pixel color $\hat{C}_f(\mathbf{r})$.

F. Optimization and Loss Function

Both networks are trained jointly by minimizing the sum of their respective mean squared errors against the ground-truth pixel color $C(\mathbf{r})$:

$$\mathcal{L} = \sum_{\mathbf{r}\in\mathcal{R}}\Big[\|\hat{C}_c(\mathbf{r}) - C(\mathbf{r})\|_2^2 + \|\hat{C}_f(\mathbf{r}) - C(\mathbf{r})\|_2^2\Big], \tag{13}$$

where $\mathcal{R}$ is the set of rays in the current mini-batch, and $\hat{C}_c, \hat{C}_f$ are the coarse and fine rendered colors, respectively. It is important that the coarse loss term is retained: even though only $\hat{C}_f$ is used at test time, training the coarse network

with supervision ensures that its weights $\{w_i\}$ are meaningful, which in turn ensures that the PDF constructed for importance sampling is informative.

A single Adam optimizer jointly manages the parameters of both the coarse and fine networks, using a decaying learning rate from 5×10^{-4} to 5×10^{-5} over the full training run. The per-step decay factor γ is derived analytically:

$$\gamma = \left(\frac{\eta_{\text{end}}}{\eta_{\text{start}}}\right)^{1/T}, \quad \gamma \approx 0.999977, \tag{14}$$

where $T = 10{,}000$ iterations, $\eta_{\text{start}} = 5 \times 10^{-4}$, and $\eta_{\text{end}} = 5 \times 10^{-5}$. This schedule allows rapid early progress with a large learning rate, followed by fine-grained convergence near the end of training.

IV. Implementation Details

A. Dataset Loading (`loadDataset`)

We use the NeRF Blender synthetic dataset [1], which provides rendered RGBA images and camera-to-world transformation matrices stored in `transforms_{train/test/val}.json`. Each image is 800×800 pixels with an alpha channel. During loading, RGBA images are composited onto a white background:

$$\mathbf{I} = \mathbf{I}_{rgb} \cdot \alpha + (1 - \alpha) \cdot \mathbf{1}, \tag{15}$$

where $\alpha \in [0, 1]$ is the per-pixel opacity. This white-background compositing must be matched exactly by the renderer so that the training signal is consistent.

B. Ray Generation (`PixelToRay`)

For each pixel (u, v) and camera pose, the function `PixelToRay` converts pixel coordinates into a world-space ray following Eqs. (2)–(3), applying the corrected sign convention. The unnormalised direction in camera space is constructed as:

$$\mathbf{d}_{\text{cam}} = \left(\frac{u - c_x}{f_x},\ -\frac{v - c_y}{f_y},\ -1.0\right), \tag{16}$$

where the negation of the y-component accounts for the image convention in which row index v increases downward while the camera's Y-axis points upward, and $z = -1.0$ reflects the Blender synthetic dataset convention that the camera looks along $-Z$ in camera space. This direction is normalised before rotation into world coordinates via the 3×3 rotation sub-block of the pose matrix, and renormalised afterward to guard against numerical drift. The ray origin is the 3×1 translation column of the pose matrix.

C. Ray Batching (`generateBatch`)

At each training iteration, a batch of $N = 4096$ rays is constructed by uniformly sampling random pixels from random training images. The pixel colors from these images serve as the ground-truth supervision signals. This stochastic procedure is critical as it effectively shuffles the dataset at every iteration, preventing the network from memorizing the order of training images.

D. Volume Rendering (`render_rays`)

The core rendering function `render_rays` accepts a network, a batch of ray origins and directions, and a tensor of t-values of shape (N, S) where S is the number of samples per ray. It:

1) Computes 3D sample positions: $\mathbf{p}_{i,j} = \mathbf{o}_i + t_{i,j}\,\mathbf{d}_i$ for ray i and sample j.
2) Reshapes to $(N \cdot S, 3)$ and queries the MLP in a single batched forward pass, recovering colors $(N, S, 3)$ and densities (N, S). The density values are taken directly from the model output, which already enforces non-negativity via the ReLU activation in the density head.
3) Computes $\delta_{i,j} = t_{i,j+1} - t_{i,j}$, scaled by the norm of the ray direction, with $\delta_{i,S} = 10^{10}$ for the last sample to approximate an infinite terminal interval.
4) Converts density to per-sample opacity via $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$, clamped to $[0, 1]$, and computes the accumulated transmittance T_i using `torch.cumprod` with a numerical floor of 10^{-10} to prevent degenerate logarithms.
5) Computes rendering weights $w_i = T_i \alpha_i$ and accumulates the composite color as $\hat{C} = \sum_i w_i \mathbf{c}_i$.
6) Applies white-background compositing to match the ground-truth preprocessing: the accumulated opacity $\text{acc} = \sum_i w_i$ is used to blend in white for rays that pass through empty space, via $\hat{C} \mathrel{+}= (1 - \text{acc}) \cdot \mathbf{1}$.

E. Hierarchical Rendering (`render` and `sample_pdf`)

The coarse stratified samples are generated by first computing the midpoints of $N_c = 64$ equally spaced bins across $[t_n, t_f]$, then constructing per-bin lower and upper bounds, and finally perturbing each bin with a uniform random offset during training or a fixed midpoint offset at test time:

```
mids  = 0.5 * (z_vals[:, :-1] + z_vals[:, 1:])
lower = torch.cat([z_vals[:, :1], mids], dim=-1)
upper = torch.cat([mids, z_vals[:, -1:]], dim=-1)
if model_coarse.training:
    t_rand = torch.rand_like(z_vals)
else:
    t_rand = torch.full_like(z_vals, 0.5)
z_vals_coarse = lower + (upper - lower) * t_rand
```

Listing 1. Stratified jitter applied to coarse samples (`render`).

This ensures that samples are spread across the full depth range during training, while test-time rendering uses deterministic midpoint samples for reproducibility.

The inverse CDF sampling is implemented in `sample_pdf`. The boundary weights at each end of the coarse weight vector are discarded and the remaining interior weights are used as the basis for the PDF, with a small epsilon $\epsilon = 10^{-5}$ added to prevent zero-probability bins from producing degenerate samples. The corresponding z-value midpoints are used as the bin centers. The CDF is built by cumulative summation, a zero is prepended to anchor the CDF at the origin, and `torch.searchsorted` is used to efficiently invert the CDF for all $N_f = 128$ uniform

samples simultaneously. Linear interpolation within each bin provides sub-bin depth resolution.

The `render` function orchestrates the full two-pass pipeline. The coarse weights from `render_rays` are fed to `sample_pdf`, which draws $N_f = 128$ importance samples. These are concatenated with the original $N_c = 64$ coarse samples, sorted in ascending depth order to form $N_c + N_f = 192$ combined samples, and passed to the fine network to produce the final rendered color. Both the coarse and fine rendered colors are returned for the joint loss computation.

F. Training Loop (`train`)

A single Adam optimizer jointly manages the parameters of both the coarse and fine networks:

```
optimizer = torch.optim.Adam(
    list(model_coarse.parameters()) +
    list(model_fine.parameters()),
    lr=args.lrate,
)
```

Listing 2. Joint optimizer over both networks (`train`).

The learning rate is decayed exponentially from 5×10^{-4} to 5×10^{-5} over $T = 10{,}000$ iterations using `torch.optim.lr_scheduler.ExponentialLR`, with the per-step factor γ computed analytically per Eq. (14). Both networks are initialized with random weights and trained jointly from scratch, or optionally resumed from the latest available checkpoint. At each iteration, a batch of 4096 rays is sampled, the hierarchical render pass is executed, and the joint coarse-plus-fine MSE loss is backpropagated through both networks in a single optimizer step followed by a scheduler step. Checkpoints are saved every 1 000 iterations (`--save_ckpt_iter` default), storing the state dictionaries of both networks, the optimizer, and the learning rate scheduler, enabling seamless training resumption from any saved iteration.

G. Testing and Evaluation (`test`)

At test time, the fine network alone determines the output color. All $H \times W$ pixels of each test image are organized into a grid of rays, which are rendered in mini-batches of 4096 rays to avoid GPU memory overflow. Crucially, the coarse stratified sampling switches from stochastic perturbation to deterministic midpoint placement (i.e., $t_{\text{rand}} = 0.5$ in each bin), ensuring that test renders are fully reproducible. The PSNR and SSIM metrics are computed per image using OpenCV (`cv2.PSNR`) and scikit-image (`structural_similarity`) respectively, then averaged over the test set. Rendered frames are assembled into an animated GIF for qualitative evaluation of temporal consistency across viewpoints.

V. Evaluation and Results

A. Experimental Setup

We evaluated our implementation on two scenes from the NeRF Blender synthetic dataset [1]: *Lego* and *Ship*. Each scene contains posed RGB(A) images rendered at 800×800 resolution, along with the corresponding camera-to-world transformations stored in the `transforms_{train,test,val}.json` files. The dataset is well suited for evaluating NeRF because it contains full 360° camera coverage and includes challenging appearance effects such as view-dependent highlights, thin structures, and fine geometric detail. Our implementation follows the hierarchical NeRF formulation from the original paper. For each ray, the coarse network is queried at $N_c = 64$ stratified samples between the near and far bounds $t_n = 2.0$ and $t_f = 6.0$. The resulting rendering weights are then used to construct a piecewise-constant PDF, from which $N_f = 128$ additional fine samples are drawn using inverse transform sampling. The union of the coarse and fine samples is sorted and passed through the fine network to produce the final rendered pixel color. Both the coarse and fine outputs are supervised during training using the sum of their mean squared errors with respect to the ground-truth pixel color, consistent with Eq. (13). Training was performed with a batch size of 4096 rays using the Adam optimizer with a learning rate initialized at 5×10^{-4} and decayed exponentially to 5×10^{-5}. Inference was performed on the test poses by rendering all image pixels in ray batches and reconstructing the full image from the fine network outputs. In addition to quantitative metrics, we generated novel-view render sequences and animated GIFs for both datasets.

B. Novel View Synthesis Results on Both Datasets

We rendered full test-time view sequences for both the *Lego* and *Ship* scenes and assembled the output frames into animated GIFs. These sequences demonstrate that the trained model learns a coherent 3D representation of the scene rather than memorizing isolated images: as the viewpoint moves around the object, the rendered geometry and appearance evolve smoothly and consistently. For the *Lego* scene, the model reliably reconstructs the overall structure of the object, including the broad silhouette, major shape components, and many of the smaller geometric details such as protrusions, support bars, and wheel structures. The rendered views also exhibit meaningful viewpoint-dependent appearance, particularly on the glossy yellow plastic surfaces, which is a direct consequence of the view-dependent color branch in the MLP architecture. For the *Ship* scene, the learned radiance field recovers the main global structure and overall scene layout, though the reconstruction exhibits a noticeably smoother and less detailed appearance compared to the *Lego* results. This discrepancy is attributable primarily to the number of training iterations used in our experiments: whereas the original paper recommends 100,000 to 300,000 iterations to reach full convergence, our training budget was substantially more constrained. The *Ship* scene, which contains large smooth hull surfaces as well as fine mast structures and rigging, is particularly sensitive to this limitation, as the fine network requires a greater number of optimization steps to resolve thin geometry and high-frequency specular appearance. Additionally, the number of fine importance samples $N_f = 128$ per ray,

while consistent with the original paper, may be insufficient to fully capture the geometry of the masts and rigging given the reduced training budget; increasing N_f in conjunction with a longer training schedule would likely yield sharper ship reconstructions.

C. Quantitative Evaluation: PSNR and SSIM

We evaluated rendering quality using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). For each test image, the rendered output from the fine network was compared to the corresponding held-out ground-truth image, with PSNR computed via OpenCV and SSIM via scikit-image, both averaged over the full test set. The averaged results for both datasets are reported in Table I. The *Lego* scene achieved a mean PSNR of 26.11 dB and a mean SSIM of 0.8796, indicating strong structural fidelity and accurate color reproduction across the test viewpoints. At the level of individual frames, representative test images yielded PSNR values spanning approximately 22.7 dB to 25.4 dB with SSIM consistently near 0.855, confirming stable reconstruction quality across diverse viewing angles. These values are particularly encouraging given that the model was trained well below the iteration count recommended by Mildenhall et al. The *Ship* scene achieved a mean PSNR of 23.73 dB and a mean SSIM of 0.7691, both of which are meaningfully lower than the corresponding *Lego* figures. The gap between the two scenes is consistent with the qualitative observation that the *Ship* renders exhibit greater surface smoothness and reduced fine detail: the lower SSIM in particular reflects a loss of structural similarity in high-frequency regions such as the rigging and hull edges, where the network has not yet converged to a sharp volumetric boundary. Both scores nonetheless confirm that the implementation is correctly learning scene geometry and view-dependent appearance, and would be expected to improve substantially with extended training.

TABLE I
AVERAGE PSNR AND SSIM ON THE TEST SET.

Dataset	PSNR (dB) ↑	SSIM ↑
Lego	26.1071	0.8796
Ship	23.7305	0.7691

D. Comparison with Ground Truth: Good and Bad Examples

Our evaluation includes qualitative side-by-side comparisons between rendered outputs and their corresponding held-out target images, following the spirit of Figs. 4 and 5 in the original NeRF paper [1]. The good examples correspond to views where the scene structure is well constrained by nearby training viewpoints and where the appearance is dominated by large smooth surfaces. In these cases, the model reconstructs object boundaries accurately, predicts plausible shading and color, and preserves visual consistency across viewpoints. On the *Lego* scene, good examples typically include frontal or moderately oblique views where the main body and wheels are clearly visible and the geometric complexity is manageable for the model at the given training budget. On the *Ship* scene, good examples are those where the dominant scene structure consists of large, well-sampled planar surfaces with limited ambiguity from thin or highly reflective regions.

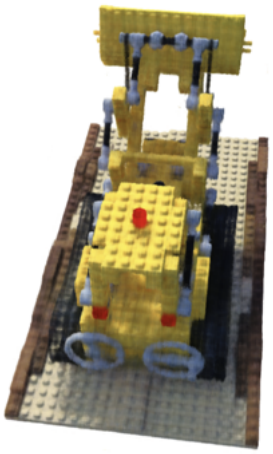

Fig. 2. Good example for the Lego dataset.

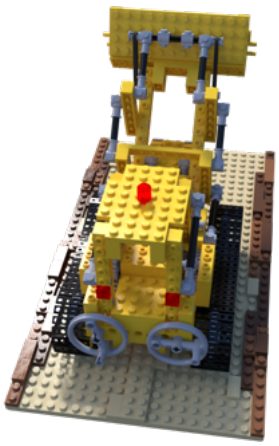

Fig. 3. Ground truth for the Lego dataset.

Fig. 4. Good example for the Ship dataset.

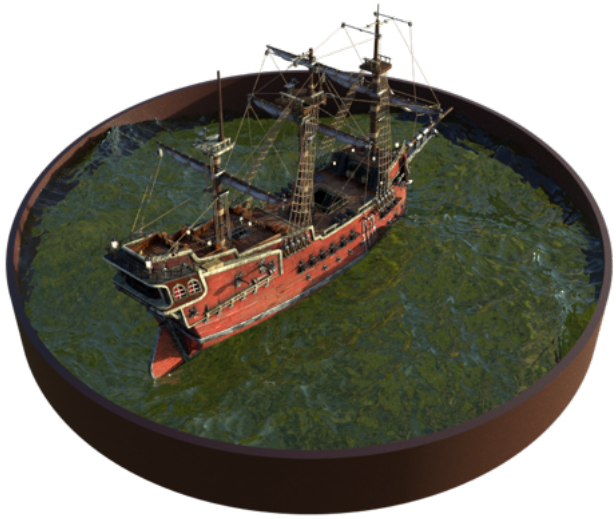

Fig. 5. Ground truth for the Ship dataset.

The bad examples tend to occur in views with thin structures, high-frequency detail, or stronger view-dependent effects. On the *Ship* scene, failure cases are particularly pronounced around the mast, rigging, and bow, where fine geometry requires a high density of well-placed importance samples and a sufficient number of gradient updates to resolve. On the *Lego* scene, failure cases are more localized, typically appearing as softened boundaries on small connector pieces or slightly incorrect specular highlights on curved plastic surfaces. These failure cases are not unexpected: they are a direct consequence of training below the convergence threshold suggested by the original paper, and they underscore the sensitivity of NeRF's hierarchical sampling to the quality of the coarse network's weight estimates. Such comparisons nonetheless serve a constructive purpose, as they make clear that the method is genuinely attempting to learn a continuous volumetric function rather than simply interpolating between training images.

Fig. 6. Bad example for the Lego dataset.

Fig. 7. Ground truth for the bad example of Lego dataset.

Fig. 8. Bad example 2 for the Lego dataset.

Fig. 9. Ground truth for the bad example 2 of Lego dataset.

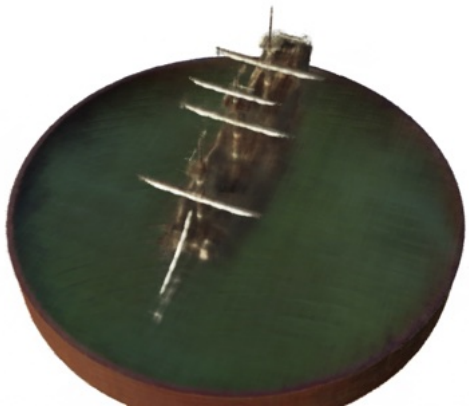

Fig. 10. Bad example for the ship dataset.

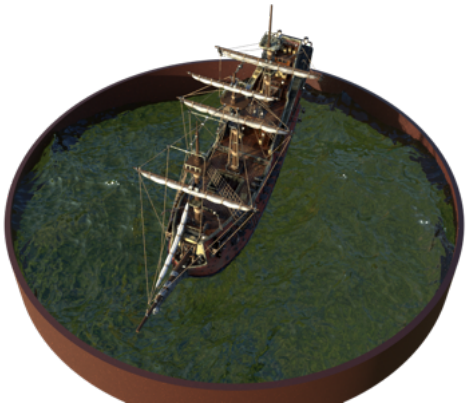

Fig. 11. Ground truth for the bad example of the ship dataset.

Fig. 12. Bad example 2 of the ship dataset.

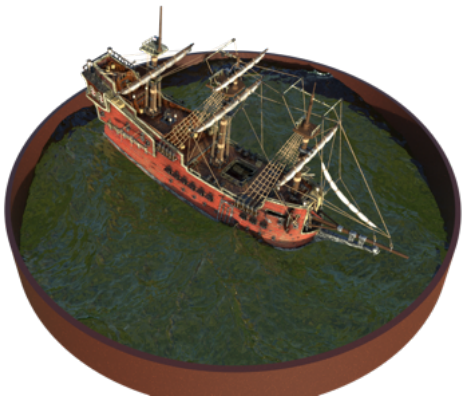

Fig. 13. Ground truth for bad example 2 of the ship dataset.

E. Comparison With and Without Positional Encoding

We performed an ablation study on the *Ship* scene to evaluate the contribution of positional encoding. The baseline configuration uses the standard Fourier feature mapping with $L = 10$ frequency bands for 3D point coordinates and $L = 4$ frequency bands for viewing directions, while the ablated configuration feeds the raw 3D coordinates and direction components directly into the MLP without any frequency expansion. The quantitative results are summarized in Table II, and the qualitative difference is both immediate and significant. Without positional encoding, the network fails to produce any meaningful scene reconstruction: throughout training, the fine network loss remains essentially stagnant at approximately 0.29–0.30, while only the coarse network loss exhibits any meaningful decrease. The consequence at test time is that all rendered images collapse to near-uniform white outputs, reflecting the fact that the fine network, receiving no informative gradient signal, converges to predicting the mean background color of the white-composited training images. This failure mode is particularly instructive because it reveals a critical dependency in the hierarchical pipeline: the fine network relies on the coarse network's weight distribution to identify where to place importance samples, but if neither network can represent high-frequency scene structure, the PDF constructed from the coarse weights is essentially flat and uninformative, depriving the fine network of the spatial guidance it needs to learn. The PSNR of 5.88 dB achieved without positional encoding is consistent with the rendered images being close to uniform white, while the SSIM of 0.6130 reflects only broad low-frequency agreement in luminance with the white-background ground-truth images rather than any meaningful structural reconstruction. Reinstating positional encoding raises the mean PSNR to 23.73 dB and the mean SSIM to 0.7691, a dramatic improvement that strongly confirms the central role of frequency expansion in enabling high-fidelity volumetric scene learning.

TABLE II
EFFECT OF POSITIONAL ENCODING ON THE LEGO SCENE.

Configuration	PSNR (dB) ↑	SSIM ↑
Without positional encoding	5.8838	0.6130
With positional encoding	23.7305	0.7691

Furthermore, we observed that appending the raw input coordinates to the positional encoding vector is equally fundamental to learning a sound scene representation. When the raw coordinates are omitted and only the sinusoidal frequency features are passed to the MLP, the network loses the low-frequency anchor that allows it to localize scene content in absolute space, producing renderings that are severely over-smoothed and structurally incoherent. This is illustrated in Fig. 14, which shows the first rendered test frame of the *Ship* scene obtained under this ablated configuration: the output lacks any discernible geometric structure, confirming that the raw coordinates and their high-frequency Fourier expansions together form a complementary representation in which neither component alone is sufficient.

Fig. 14. Rendered test frame 0 of the Ship scene when raw coordinates are excluded from the positional encoding input, illustrating the collapse of geometric structure that results from the absence of a low-frequency spatial anchor.

F. Issues Encountered and How They Were Resolved

Several non-trivial implementation issues arose during development. The first concerned the camera-ray convention required by the Blender synthetic dataset. This dataset adopts a convention in which the camera looks along the negative z axis in camera space and image row coordinates increase downward, which is the opposite of several common computer vision conventions. An incorrect sign assignment in the ray construction step caused rays to be cast into the wrong hemisphere, preventing any meaningful scene reconstruction from the outset. This was resolved by explicitly negating both the y-component and the z-component of the camera-space ray direction before rotating into world coordinates, matching the dataset convention exactly. The second issue arose in the hierarchical rendering pipeline, specifically in the construction of the fine importance samples. Earlier versions of the implementation produced unstable or uninformative fine samples due to numerical errors in the PDF normalization and CDF inversion steps, causing the fine network to receive poorly distributed samples concentrated away from the actual scene surfaces. Revising the inverse-CDF sampling procedure, adding a small epsilon floor to prevent degenerate zero-probability bins, and ensuring that fine samples were correctly merged with and sorted alongside the coarse samples substantially improved rendering stability and reconstruction quality. The third issue concerned consistency between the dataset preprocessing and the renderer. Since the RGBA training images are composited onto a white background during loading, the rendered rays must apply the identical white-background blending operation; failing to do so creates a systematic mismatch between the target distribution and the renderer's output that the network cannot overcome regardless

of capacity or training duration. Enforcing this consistency resolved the mismatch and stabilized training convergence. Taken together, these issues illustrate a broader lesson: NeRF's correctness depends as much on the fidelity of the sampling and rendering pipeline as on the MLP architecture itself.

VI. Discussion

A. Architecture and Pipeline Insights

One of the most important insights from this project is that NeRF should be understood as a complete differentiable rendering pipeline rather than merely an MLP that predicts color. The full system consists of several tightly coupled components, including camera ray generation, positional encoding, two separate MLPs, hierarchical sampling, differentiable alpha compositing, and image-space reconstruction loss, and the success of the method depends on all of these components operating correctly and consistently. The coarse and fine networks play architecturally identical but functionally distinct roles in this pipeline. The coarse network is responsible for generating a rough but spatially coherent estimate of scene density along each ray; its rendering weights are then normalized into a probability density function that guides the allocation of the fine network's samples. The fine network, in turn, uses this distribution to concentrate its representational capacity on the regions of each ray most likely to contain visible scene content, producing the final rendered color from the union of coarse and fine samples. This two-stage strategy provides an elegant trade-off between computational efficiency and rendering fidelity that would be difficult to achieve with either purely uniform or purely coarse sampling. The ablation studies conducted in this work further highlighted how individual architectural decisions can make the difference between a functioning system and a complete failure, underscoring that the components of the pipeline are not independently interchangeable.

B. Strengths of the Method

The primary strength of NeRF lies in its ability to represent a complex three-dimensional scene as a continuous implicit function, side-stepping the memory scaling limitations of voxel grids and the topological restrictions of surface-based representations. Because the color output depends jointly on spatial position and viewing direction, the model can capture non-Lambertian appearance effects such as specular highlights and iridescence that are fundamentally inaccessible to purely geometric representations. A second major strength is that the entire system is trainable from posed two-dimensional images alone, without any form of explicit 3D supervision. The use of differentiable volume rendering allows the geometry and appearance of the scene to be inferred implicitly by requiring a single radiance field to explain the full set of training observations simultaneously. The ablation experiments conducted in this work further confirm two of the central claims of the original paper: first, positional encoding is not a peripheral enhancement but a foundational mechanism whose removal causes the fine network to cease learning entirely and reduces all rendered outputs to undifferentiated white images; and second, the raw input coordinates must be retained alongside their sinusoidal frequency expansions, as they provide the low-frequency spatial anchor without which the network cannot localize scene content in absolute space, regardless of how many high-frequency Fourier features are supplied.

C. Limitations and Failure Cases

The most practically significant limitation observed in our experiments is the sensitivity of reconstruction quality to training duration. The original paper reports that 100,000 to 300,000 iterations are required to reach the PSNR values of 28–32 dB reported for the Blender synthetic scenes, a training budget that was not achievable within the constraints of this project. The consequence is most visible in the *Ship* scene, where the rendered outputs exhibit excessive smoothness and fail to resolve thin geometric structures such as masts and rigging. This smoothness is not merely a qualitative limitation: it is reflected quantitatively in the lower SSIM score of 0.7691 for the *Ship* relative to 0.8796 for the *Lego*, confirming that structural detail is being lost. Extending the training schedule toward the paper's recommended range, and potentially increasing the number of fine importance samples N_f beyond 128 to improve surface localization for geometrically thin regions, would both be expected to yield sharper reconstructions. A second limitation, revealed through our ablation study, is that the positional encoding must be constructed with care: omitting the raw coordinates from the encoding input, even while retaining the full set of sinusoidal features, causes geometric structure to collapse entirely in the rendered outputs. This finding suggests that the low-frequency component provided by the raw coordinates and the high-frequency components provided by the Fourier expansion play complementary and non-substitutable roles in the network's ability to represent scene geometry. A third limitation is the computational cost of inference, which requires evaluating the MLP at $N_c + N_f = 192$ points per ray for every pixel in the output image, making the generation of full novel-view animations costly even after training is complete. Finally, vanilla NeRF assumes a static scene with known camera extrinsics and does not generalize across scenes or handle dynamic objects, which motivates the large body of subsequent work on accelerated, generalizable, and dynamic NeRF variants.

VII. Conclusion

In this project, we implemented the original NeRF framework for novel view synthesis from posed RGB images, reproducing the full pipeline described in Mildenhall et al. [1]: ray generation from camera intrinsics and poses, Fourier positional encoding of spatial coordinates and viewing directions, an 8-layer MLP with skip connection and view-dependent color branch, differentiable volume rendering via discrete alpha compositing, and a hierarchical coarse-to-fine sampling strategy using two independently parameterized networks. The coarse network generates an initial radiance field estimate

whose rendering weights drive PDF-based importance sampling, while the fine network refines the final rendered color over the combined set of 192 samples per ray.

The evaluation demonstrated that the implementation successfully learns coherent volumetric scene representations from posed images alone. On the *Lego* scene, the model achieved a mean PSNR of 26.11 dB and a mean SSIM of 0.8796, with individual test frames reaching up to 25.43 dB, confirming that the hierarchical pipeline correctly learns both geometry and view-dependent appearance at the given training budget. The *Ship* scene achieved a mean PSNR of 23.73 dB and a mean SSIM of 0.7691; the qualitatively smoother appearance of the ship renders is consistent with this quantitative gap and is directly attributable to insufficient training iterations relative to the 100,000 to 300,000 iterations recommended by the original paper. Increasing the number of fine importance samples and extending training toward the recommended range would be the most direct path to closing this gap for geometrically complex or thin-structured scenes. The positional encoding ablations produced the most striking results of the evaluation. Removing the encoding entirely caused the fine network loss to plateau at approximately 0.29–0.30 throughout training while only the coarse loss continued to decrease, with all test renders collapsing to uniform white images and a PSNR of only 5.88 dB. A second ablation revealed that retaining the sinusoidal frequency features while omitting the raw input coordinates is equally damaging: without the low-frequency spatial anchor that the raw coordinates provide, the network loses the ability to localize scene content in absolute space and again produces structurally incoherent outputs. Together, these two ablations establish that the full positional encoding combining raw coordinates and their Fourier expansions is a structural prerequisite for the system to function, not a tunable hyperparameter.

A central practical lesson of this project is that NeRF's correctness depends as much on the fidelity of the rendering pipeline as on the network architecture. Errors in camera ray convention, positional encoding construction, inverse-CDF sampling, or background compositing are not recoverable through additional training capacity and must be addressed at the implementation level. Debugging and resolving these issues was essential to obtaining the working system reported here. Taken together, the quantitative results, qualitative comparisons, and ablation studies confirm both the theoretical elegance and the engineering precision that underlie neural radiance fields, and provide a clear picture of the conditions under which the method succeeds and the directions in which further improvement is achievable.

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